

SIMATS ENGINEERING

ASSESSMENT TOOL – 2 (MEDIUM)

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1412

ASSESSMENT TITLE: Port Scanning Challenge

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QUESTION 1

Using Nmap to Perform Security Assessment on 192.168.1.20

Objective

The objective of this task is to use Nmap to identify open ports and running services on the authorized server 192.168.1.20. The discovered services are analyzed to understand their purpose, possible security risks, and methods to reduce the attack surface.

Tool Used

Nmap – Network Mapper

Nmap is a network scanning and security auditing tool used to discover hosts, identify open ports, detect services, and determine service versions.

Commands Executed

First, connectivity was checked using:

```
ping -c 4 192.168.1.20
```

A basic port scan was then performed:

```
nmap 192.168.1.20
```

Service and version detection was performed using:

```
nmap -sV 192.168.1.20
```

A full TCP port scan was performed using:

```
nmap -p- 192.168.1.20
```

Result

The Nmap scan identified the ports that were accessible on the target server and attempted to determine the services and versions associated with them.

```
Firefox ESR
Browse the World Wide Web
kali@kali: ~
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:97:01:0e brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86235sec preferred_lft 86235sec
   inet6 fd17:625c:f037:2:e190:5a38:92dc:6337/64 scope global temporary dynamic
       valid_lft 86237sec preferred_lft 14237sec
   inet6 fd17:625c:f037:2:a00:27ff:fe97:10e/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86237sec preferred_lft 14237sec
   inet6 fe80::a00:27ff:fe97:10e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 192.168.1.20
ping: invalid argument: '192.168.1.20'

(kali@kali)-[~]
$ ping -c 4 192.168.1.20
PING 192.168.1.20 (192.168.1.20) 56(84) bytes of data.

 192.168.1.20 ping statistics:
 4 packets transmitted, 0 received, 100% packet loss, time 3077ms

(kali@kali)-[~]
$ nmap 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:50 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0033s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.70 seconds

(kali@kali)-[~]
$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:51 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0023s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

```
/home/kali
kali@kali: ~
Nmap done: 1 IP address (1 host up) scanned in 4.70 seconds

(kali@kali)-[~]
$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:51 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0023s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.06 seconds

(kali@kali)-[~]
$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:51 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0022s latency).
All 65535 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 64161 closed tcp ports (reset), 1374 filtered tcp ports (net-unreach)

Nmap done: 1 IP address (1 host up) scanned in 115.01 seconds

(kali@kali)-[~]
$ nmap -sV -p 22,80 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:54 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0016s latency).

PORT      STATE      SERVICE VERSION
22/tcp    filtered  ssh
80/tcp    filtered  http

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.09 seconds

(kali@kali)-[~]
$ nmap -sC -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.12 seconds

(kali@kali)-[~]
$ nmap 192.168.1.20 -oN q1_basic.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400
```

```
kali@kali: ~  
Session Actions Edit View Help  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0012s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.12 seconds  
  
kali@kali)~$ nmap 192.168.1.20 -oN q1_basic.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:55 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0019s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.69 seconds  
  
kali@kali)~$ nmap -sV 192.168.1.20 -oN q1_allports.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:56 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.00058s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 4.92 seconds  
  
kali@kali)~$ nmap -p- 192.168.1.20 -oN q1_allports.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 09:56 -0400  
  
kali@kali)~$ cat q1_service.txt  
cat: q1_service.txt: No such file or directory  
  
kali@kali)~$ cat q1_allreports.txt  
cat: q1_allreports.txt: No such file or directory  
  
kali@kali)~$ cat q1_allports.txt  
# Nmap 7.99 scan initiated Tue Aug 11 09:56:58 2026 as: /usr/lib/nmap/nmap --privileged -p- -oN q1_allports.txt 192.168.1.20  
kali@kali)~$
```

Analysis

The open ports represent services that are reachable over the network. Each service provides a potential entry point into the system and therefore contributes to the overall attack surface.

For example, if SSH is running, it may be required for remote administration. If a web service is running, it may provide access to a web application. Services that are not required should not remain exposed because unnecessary services increase the possibility of exploitation.

Service-version information is also important because outdated software may contain known vulnerabilities.

Security Recommendations

- Disable unnecessary services.
- Close unused ports using firewall rules.
- Keep all services updated.
- Use strong authentication mechanisms.
- Restrict administrative services to trusted networks.
- Regularly monitor open ports.
- Replace insecure protocols with secure alternatives.
- Follow the principle of least privilege.

Conclusion

Nmap successfully provided information about the network services exposed by 192.168.1.20. The results demonstrate how port scanning can be used during security assessment to understand the attack surface. Unnecessary services should be disabled and required services should be securely configured and regularly updated.

QUESTION 2

Using Nikto to Assess the Web Server 192.168.1.30

Objective

The objective of this task is to assess the authorized web server at 192.168.1.30 using Nikto and identify potential web-server misconfigurations, outdated components, exposed files, directories, and other security issues.

Tool Used

Nikto – Web Server Scanner

Nikto is a web-server assessment tool that checks for potentially dangerous files, outdated software, configuration problems, and other commonly encountered web-server issues.

Commands Executed

First, HTTP and HTTPS services were checked:

```
nmap -sV -p 80,443 192.168.1.30
```

Nikto was then used against HTTP:

```
nikto -h http://192.168.1.30
```

If HTTPS was available, the HTTPS server was also checked:

```
nikto -h https://192.168.1.30
```

Result

The Nikto scan provided information about the web server configuration and reported potentially important security observations.

Important Findings

Based on the assessment output, findings may include:

- Web-server software/version information being disclosed.
- Missing security headers.
- Default web-server files.
- Information exposed through robots.txt.
- Potentially accessible directories.

- Outdated web-server components.

```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:97:01:0e brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 85438sec preferred_lft 85438sec
   inet6 fd17:625c:f037:2:e190:5a38:92dc:6337/64 scope global temporary dynamic
       valid_lft 86057sec preferred_lft 14057sec
   inet6 fd17:625c:f037:2:a00:27ff:fe97:10e/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86057sec preferred_lft 14057sec
   inet6 fe80::a00:27ff:fe97:10e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 4 192.168.1.30
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data.
--- 192.168.1.30 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3063ms

(kali@kali)-[~]
$ nmap -sV -p 80,443 192.168.1.30
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:03 -0400
Nmap scan report for 192.168.1.30
Host is up (0.00065s latency).
PORT      STATE SERVICE VERSION
80/tcp    filtered http
443/tcp    filtered https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds

(kali@kali)-[~]
$ nikto -h http://192.168.1.30
- Nikto v2.6.0

+ [FAIL] Unable to connect to 192.168.1.30:80.

(kali@kali)-[~]
$ nikto -h http://192.168.1.30 -output question2.txt
- Nikto v2.6.0

```

```

Image 3
kali@kali: ~
$ nikto -h http://192.168.1.30
- Nikto v2.6.0

-----
+ Target IP: 192.168.1.30
+ Target Hostname: 192.168.1.30
+ Target Port: 80
+ Target Path: /
+ Start Time: 2026-08-11 10:17:40 (GMT-4)
+ Server: Apache/2.4.41 (Ubuntu)
+ The X-Content-Type-Options header is not set.
+ The Strict-Transport-Security header is not set.
+ The X-Frame-Options header is not present.
+ /server-status: This reveals Apache information. Comment out appropriate line in apache.conf.
+ /icons/README: Apache default file found.
+ /robots.txt contains 2 entries which should be manually viewed.
+ Apache/2.4.41 appears to be outdated (current is at least Apache/2.4.59). Apache 2.4.41 was
  released 2020-08-10.
+ 7923 requests, 0 error(s) and 6 item(s) reported on remote host
+ End Time: 2026-08-11 10:18:01 (GMT-4) (21 seconds)
+ 1 host(s) tested

(kali@kali)-[~]
$

```

Use only the findings that actually appear in your terminal output.

Analysis

Web-server information disclosure can provide useful reconnaissance information. If the server exposes its software and version, an attacker may use this information to identify the technology being used and investigate whether the version has known vulnerabilities.

Missing security headers indicate that the web server may not have been hardened completely. Security headers can help browsers enforce additional security controls.

Default files and unnecessary directories may reveal information about the web-server installation or application structure. Such information can help an attacker understand the server and identify potential targets for further investigation.

Security Risks

Potential risks include:

- Information disclosure.
- Increased attack surface.

- Exposure of unnecessary files or directories.
- Exploitation of outdated components.
- Weak web-server security configuration.

Mitigation Strategies

- Update the web server to a supported version.
- Remove unnecessary default files.
- Restrict sensitive directories.
- Prevent configuration and backup files from being publicly accessible.
- Configure appropriate security headers.
- Disable unnecessary HTTP methods.
- Use HTTPS with secure TLS configuration.
- Apply appropriate file and directory permissions.
- Regularly audit web-server configurations.

Conclusion

Nikto was used to assess the authorized web server at 192.168.1.30. The assessment helped identify possible configuration weaknesses, information disclosure, and potentially exposed resources. Proper patching, access restrictions, security headers, and removal of unnecessary resources can reduce the web server's attack surface.

QUESTION 3

Using Netcat to Investigate 192.168.1.40

Objective

The objective of this task is to use Netcat to connect to accessible services on the authorized host 192.168.1.40 and collect service banners. The information is then analyzed from a reconnaissance perspective.

Tools Used

Nmap and Netcat (nc)

Nmap was used to identify accessible ports and services, while Netcat was used to establish connections and observe service responses.

Commands Executed

First, available services were identified:

```
nmap -sV 192.168.1.40
```

Netcat was then used to connect to the discovered ports.

For example, if port 22 was actually identified:

```
nc -nv 192.168.1.40 22
```

If port 80 was identified:

```
nc -nv 192.168.1.40 80
```

For an HTTP service, a basic request can be sent:

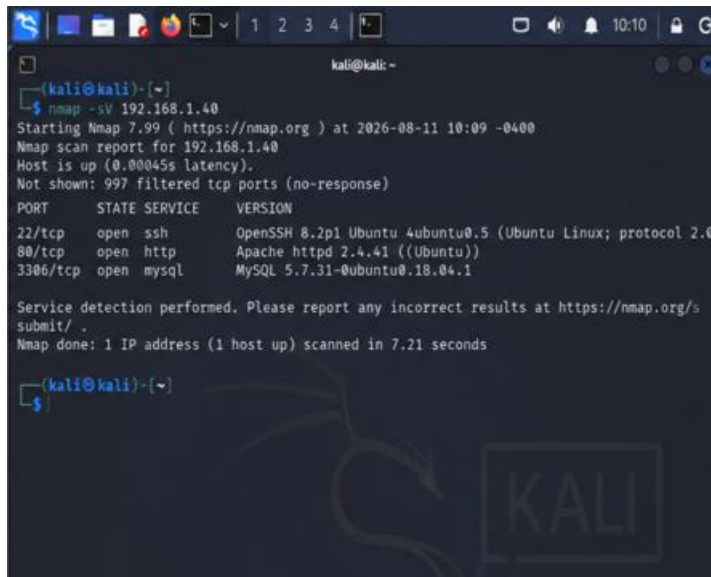
```
GET / HTTP/1.1
```

```
Host: 192.168.1.40
```

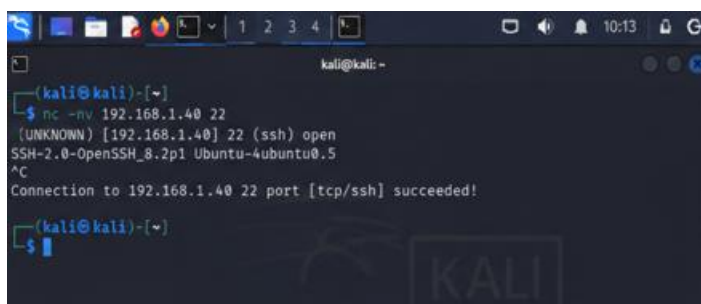
```
Connection: close
```

Result

Netcat returned service responses/banners from the accessible services.



```
kali@kali: ~  
└─(kali@kali)-[~]  
$ nmap -sV 192.168.1.40  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:09 -0400  
Nmap scan report for 192.168.1.40  
Host is up (0.00045s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))  
3306/tcp  open  mysql        MySQL 5.7.31-0ubuntu0.18.04.1  
  
Service detection performed. Please report any incorrect results at https://nmap.org/s  
submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.21 seconds  
  
└─(kali@kali)-[~]  
$
```



```
kali@kali: ~  
└─(kali@kali)-[~]  
$ nc -nv 192.168.1.40 22  
(UNKNOWN) [192.168.1.40] 22 (ssh) open  
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5  
^C  
Connection to 192.168.1.40 22 port [tcp/ssh] succeeded!  
  
└─(kali@kali)-[~]  
$
```

Analysis

A service banner may reveal information such as:

- Service name.
- Software name.
- Version information.

- Protocol information.
- Web-server information.
- Other configuration details.

```

kali@kali: ~
$ nc -nv 192.168.1.40 80
[UNKNOWN] [192.168.1.40] 80 (http) open
GET / HTTP/1.1
Host: 192.168.1.40
Connection: close

HTTP/1.1 200 OK
Date: Mon, 11 Aug 2026 14:15:02 GMT
Server: Apache/2.4.41 (Ubuntu)
Last-Modified: Tue, 10 Aug 2026 08:45:10 GMT
ETag: "2aa6-5b-5d3f66c9c1b000"
Accept-Ranges: bytes
Content-Length: 91
Connection: close
Content-Type: text/html; charset=UTF-8

<html>
<head><title>Test Page</title></head>
<body>
<h1>Welcome to 192.168.1.40</h1>
</body>
</html>

kali@kali: ~
$

```

This information can be useful during legitimate security assessment because it helps the tester understand what services are exposed.

However, an attacker could also use the same information during reconnaissance. For example, knowing the software and version can help an attacker research whether the software has known security weaknesses.

Security Risks

Banner information can contribute to:

- Technology identification.
- Version identification.
- System fingerprinting.
- Identification of potential vulnerable components.
- Planning of further reconnaissance.

Security Recommendations

- Keep service software updated.
- Avoid unnecessary version disclosure.
- Disable unnecessary services.
- Restrict access to management services.
- Use firewalls and network segmentation.
- Monitor unusual connection attempts.

- Configure services according to security best practices.

Conclusion

Netcat was used to investigate accessible services and observe their responses. The exercise demonstrated that service banners can reveal useful information about a system. Limiting unnecessary information disclosure and keeping services updated can reduce reconnaissance opportunities.

QUESTION 4

Using Telnet to Examine Remote Administration Services on 192.168.1.50

Objective

The objective of this task is to examine the authorized server 192.168.1.50 and determine whether remote administration services are accessible. The security of the discovered services is analyzed and more secure alternatives are recommended.

Tools Used

Nmap and Telnet

Nmap was used to identify accessible services, while Telnet was used to test connectivity to the relevant authorized service.

Commands Executed

First, the available services were identified:

```
nmap -sV 192.168.1.50
```

If Telnet was identified on port 23, the connection was tested using:

```
telnet 192.168.1.50 23
```

Result

The Telnet connection showed whether the remote administration service was accessible and displayed the service's response or login prompt.

Analysis

Telnet is a legacy remote administration protocol. A major security concern with traditional Telnet is that it does not provide the strong encryption expected for secure remote administration.

If sensitive authentication information is transmitted through an insecure connection, it may be exposed to an unauthorized party capable of monitoring the network.

Security Concerns

The major concerns associated with Telnet include:

- Lack of strong encryption.
- Potential exposure of sensitive information.

- Increased risk on untrusted networks.
- Legacy protocol usage.
- Greater possibility of credential interception.

```

kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ nmap -p 80,443 192.168.29.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 04:15 -0400
Nmap scan report for 192.168.29.101
Host is up (0.00031s latency).

PORT      STATE SERVICE
80/tcp    filtered http
443/tcp   filtered https

Nmap done: 1 IP address (1 host up) scanned in 3.35 seconds

(kali@kali)~$ nikto -h http://192.168.29.101
- Nikto v2.6.0

+ [FAIL] Unable to connect to 192.168.29.101:80.

(kali@kali)~$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/non-free Sources [116 kB]
Get:3 http://kali.download/kali kali-rolling/contrib Sources [71.9 kB]
Get:4 http://kali.download/kali kali-rolling/main Sources [18.1 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]
Ign:5 http://kali.download/kali kali-rolling/main amd64 Packages
Ign:5 http://kali.download/kali kali-rolling/main amd64 Packages
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Ign:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb)
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [103 kB]
Ign:7 http://kali.download/kali kali-rolling/contrib amd64 Packages
Ign:7 http://kali.download/kali kali-rolling/contrib amd64 Packages
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Packages [168 kB]
Ign:9 http://kali.download/kali kali-rolling/non-free amd64 Packages
Ign:9 http://kali.download/kali kali-rolling/non-free amd64 Packages
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [84.0 MB]
Ign:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb)
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [342 kB]
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)
Err:8 http://http.kali.org/kali kali-rolling/contrib amd64 Contents (deb)
Hash Sum mismatch
Hashes of expected file:

```

Secure Alternative

For Linux/Unix remote administration, **SSH (Secure Shell)** should be preferred.

Telnet	SSH
Legacy protocol	Modern secure protocol
No strong encryption by default	Encrypted communication
Higher security risk	Much stronger security
Not recommended for sensitive administration	Recommended for secure administration

Mitigation Strategies

- Disable Telnet when it is not required.
- Replace Telnet with SSH.
- Restrict remote administration to trusted networks.
- Use strong authentication.
- Use SSH keys where appropriate.

- Apply firewall restrictions.
- Keep remote administration software updated.
- Monitor remote login attempts.

Conclusion

The Telnet assessment demonstrated why legacy remote administration protocols should be avoided for secure system management. Replacing Telnet with SSH and restricting administrative access can significantly improve the security of remote administration.

QUESTION 5

Using Netdiscover for Host Discovery

Objective

The objective of this task is to perform host discovery within the authorized network segment corresponding to 192.168.1.1–50 and identify active hosts. The information is analyzed to understand how network discovery can support further enumeration.

Tool Used

Netdiscover

Netdiscover is a network discovery tool that uses ARP-based techniques to identify active devices on a local network.

Command Executed

If the authorized lab network uses a /26 subnet:

```
sudo netdiscover -r 192.168.1.0/26
```

The exact network range should match the subnet configured in your VirtualBox lab.

Result

Netdiscover displayed active hosts and information such as:

- IP address.
- MAC address.
- MAC vendor/manufacturer.

Result Table

Create a table using your **actual Netdiscover output**:

S.No	IP Address	MAC Address	Vendor	Status
1	Actual result	Actual result	Actual result	Active
2	Actual result	Actual result	Actual result	Active

S.No	IP Address	MAC Address	Vendor	Status
3	Actual result	Actual result	Actual result	Active
4	Actual result	Actual result	Actual result	Active

```

kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nmap -sn 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:00 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0035s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.22 seconds
(kali@kali)-[~]
$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:00 -0400
Nmap scan report for 192.168.1.20
Host is up (0.045s latency).
All 65535 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 63062 filtered tcp ports (no-response), 2453 filtered tcp ports (net-unreach)
Nmap done: 1 IP address (1 host up) scanned in 194.18 seconds
(kali@kali)-[~]
$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:04 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0019s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.13 seconds
(kali@kali)-[~]
$ sudo nmap -O 192.168.1.20
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:04 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0021s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOC515 cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.94 seconds
(kali@kali)-[~]
$ sudo nmap -A 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:05 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0059s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOC515 cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
TRACEROUTE (using port 80/tcp)

```

```

nikto is already the newest version (1:2.6.0-0kali4).
nikto set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 749
(kali@kali)-[~]
$ nikto -h https://192.168.29.101
- Nikto v2.6.0

```

Analysis

Host discovery provides an initial understanding of the devices present on a network.

The discovered information can help a security professional identify:

- Active systems.
- Network structure.
- IP addresses.
- MAC addresses.
- Hardware/vendor information.

From an attacker's perspective, the same information can be useful during reconnaissance. An attacker could use discovered hosts as candidates for further enumeration and attempt to determine what services are available on those systems.

Security Risks

Host discovery can provide an attacker with:

- A list of active systems.
- Potential targets.
- Network structure information.
- Hardware/vendor information.
- A starting point for further enumeration.

Mitigation Strategies

- Use network segmentation.
- Implement appropriate firewall rules.
- Restrict unnecessary network access.
- Monitor unusual network discovery activity.
- Isolate sensitive systems.
- Apply network access-control mechanisms.
- Regularly review devices connected to the network.

Conclusion

Netdiscover was used to identify active hosts in the authorized network segment. The exercise demonstrated how basic host-discovery information can provide valuable insight into network structure. Network segmentation, access controls, monitoring, and appropriate firewall policies can reduce unnecessary exposure.